Functions & Transformations Error Analysis TEACHER KEY

Find the mistake. Explain it. Fix it.

Name: _____ Period: _____ Date: _____

1. Horizontal shift sign

ORIGINAL PROBLEM

The graph of $f(x) = |x|$ (dashed) is transformed to $g(x) = |x + 3| - 2$. State the vertex of g and graph it.

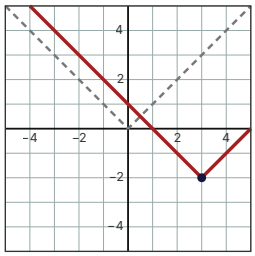
INCORRECT WORKED SOLUTION

$g(x) = |x + 3| - 2$

The +3 means move 3 units right.

The −2 means move 2 units down.

Vertex: (3, −2)



Dashed: $f(x)$. Solid: student's graph.

A. EXPLAIN THE MISTAKE (1 POINT)

Inside the function, $x + 3 = x - (-3)$, so the graph moves LEFT 3, not right. Horizontal shifts work opposite to the sign. The −2 is outside, so down 2 is right.

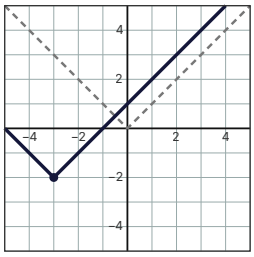
B. SHOW CORRECT WORK (1 POINT)

$g(x) = |x - (-3)| - 2$

Left 3 and down 2

Vertex: (−3, −2)

Check: $g(-3) = |0| - 2 = -2$



Dashed: $f(x)$. Solid: $g(x)$.

2. Function notation: $f(x + k)$ vs $f(x) + k$

ORIGINAL PROBLEM

Let $f(x) = x^2 - 3x$. Find and simplify $f(x + 2)$.

INCORRECT WORKED SOLUTION

$f(x + 2) = f(x) + 2$

$= x^2 - 3x + 2$

A. EXPLAIN THE MISTAKE (1 POINT)

$f(x + 2)$ means replace every x in the formula with $(x + 2)$. The 2 goes INSIDE the function, not onto the output. Test $x = 0$: $f(0 + 2) = f(2) = -2$, but the student's answer gives 2.

B. SHOW CORRECT WORK (1 POINT)

$f(x + 2) = (x + 2)^2 - 3(x + 2)$

$= x^2 + 4x + 4 - 3x - 6$

$= x^2 + x - 2$

Check $x = 0$: $f(2) = 4 - 6 = -2$ and $0 + 0 - 2 = -2$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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3. Reflections: $-f(x)$ vs $f(-x)$

ORIGINAL PROBLEM

The graph of $f(x) = \sqrt{x + 3}$ is dashed. Graph $h(x) = f(-x)$.

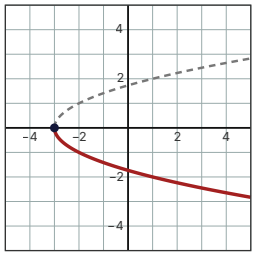
INCORRECT WORKED SOLUTION

$h(x) = f(-x)$

The negative changes the sign of every output,

so $h(x) = -\sqrt{x + 3}$.

Reflect f in the x-axis.



Dashed: $f(x)$. Solid: student's graph.

A. EXPLAIN THE MISTAKE (1 POINT)

In $f(-x)$ the negative is on the INPUT, so each x-coordinate changes sign: a reflection in the y-axis. Reflecting in the x-axis is $-f(x)$, with the negative on the output.

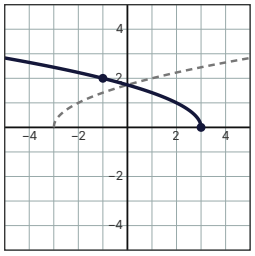
B. SHOW CORRECT WORK (1 POINT)

$h(x) = f(-x) = \sqrt{-x + 3}$

Reflect in the y-axis: $(x, y) \rightarrow (-x, y)$

$(-3, 0) \rightarrow (3, 0)$ $(-2, 1) \rightarrow (2, 1)$

$(1, 2) \rightarrow (-1, 2)$ Domain: $x \leq 3$



Dashed: $f(x)$. Solid: $h(x)$.

4. Domain: square root and denominator

ORIGINAL PROBLEM

Find the domain of $h(x) = \dfrac{\sqrt{x + 2}}{x - 4}$. Write it in interval notation.

INCORRECT WORKED SOLUTION

$x + 2 \geq 0$

$x \geq -2$

Domain: $[-2, \infty)$

A. EXPLAIN THE MISTAKE (1 POINT)

Two things restrict this function. The square root needs $x + 2 \geq 0$, but the denominator also cannot be 0: $x - 4 \neq 0$. The student checked only the radicand and left in $x = 4$.

B. SHOW CORRECT WORK (1 POINT)

Radicand: $x + 2 \geq 0$, so $x \geq -2$

Denominator: $x - 4 \neq 0$, so $x \neq 4$

Domain: $[-2, 4) \cup (4, \infty)$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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5. Horizontal compression: $f(2x)$

ORIGINAL PROBLEM

The graph of $f(x) = x^2 - 4$ is dashed. Graph $g(x) = f(2x)$ and list its x-intercepts.

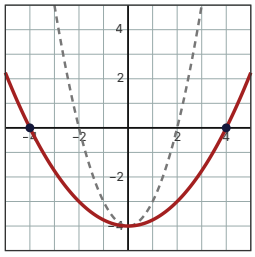
INCORRECT WORKED SOLUTION

$g(x) = f(2x)$

The 2 stretches the graph horizontally by a factor of 2,

so each x-value doubles:

$(-2, 0) \to (-4, 0)$ and $(2, 0) \to (4, 0)$



Dashed: $f(x)$. Solid: student's graph.

A. EXPLAIN THE MISTAKE (1 POINT)

A number multiplying x INSIDE the function works opposite to what you expect: $f(2x)$ compresses the graph horizontally, so each x-value is cut in half. The student doubled them.

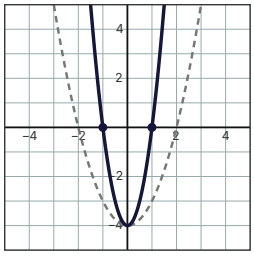
B. SHOW CORRECT WORK (1 POINT)

$g(x) = (2x)^2 - 4 = 4x^2 - 4$

$(-2, 0) \to (-1, 0)$ and $(2, 0) \to (1, 0)$

Check: $4x^2 - 4 = 0$ gives $x = \pm 1$

Vertex stays at $(0, -4)$.



Dashed: $f(x)$. Solid: $g(x)$.

6. Composition order (modeling)

ORIGINAL PROBLEM

A jacket costs \$95. A \$15 coupon is applied first, $C(p) = p - 15$, and then a 20% discount, $D(p) = 0.8p$. Find the final price using $(D \circ C)(95)$.

INCORRECT WORKED SOLUTION

$(D \circ C)(95) = C(D(95))$

$= C(0.8 \cdot 95) = C(76)$

$= 76 - 15 = \$61$

A. EXPLAIN THE MISTAKE (1 POINT)

$(D \circ C)(95)$ means $D(C(95))$: work from the INSIDE out, so C happens first. The student did D first, which is $(C \circ D)(95)$. Order matters because the coupon is worth more before the discount is taken.

B. SHOW CORRECT WORK (1 POINT)

$(D \circ C)(95) = D(C(95))$

$C(95) = 95 - 15 = 80$

$D(80) = 0.8 \cdot 80 = \$64$

Check: $(D \circ C)(p) = 0.8(p - 15) = 0.8p - 12$

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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7. Inverse notation: f^{-1} is not a reciprocal

ORIGINAL PROBLEM

Let $f(x) = 5 - 3x$. Find $f^{-1}(-7)$.

INCORRECT WORKED SOLUTION

$$f^{-1}(-7) = \frac{1}{f(-7)}$$

$$f(-7) = 5 - 3(-7) = 26$$

$$f^{-1}(-7) = \frac{1}{26}$$

A. EXPLAIN THE MISTAKE (1 POINT)

The -1 in f^{-1} is not an exponent. It means "inverse function," not reciprocal. $f^{-1}(-7)$ is the INPUT that f turns into -7. (A reciprocal would be written $\frac{1}{f(x)}$.)

B. SHOW CORRECT WORK (1 POINT)

$f^{-1}(-7) = k$ means $f(k) = -7$

$5 - 3k = -7$

$k = 4$

Check: $f(4) = 5 - 12 = -7$

8. Inverse graph: reflect in $y = x$

ORIGINAL PROBLEM

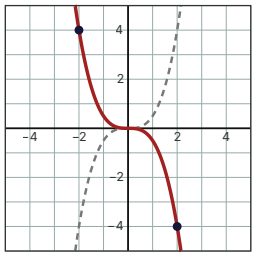
The graph of $f(x) = \frac{x^3}{2}$ is dashed. Graph $f^{-1}(x)$.

INCORRECT WORKED SOLUTION

An inverse "undoes" f, so it flips the outputs:

$(x, y) \to (x, -y)$, so $(2, 4) \to (2, -4)$

$$f^{-1}(x) = -\frac{x^3}{2}$$



Dashed: $f(x)$. Solid: student's graph.

A. EXPLAIN THE MISTAKE (1 POINT)

An inverse swaps inputs and outputs: $(x, y) \to (y, x)$, a reflection in the line $y = x$. Flipping only the outputs reflects in the x-axis, which gives $-f(x)$, not $f^{-1}(x)$.

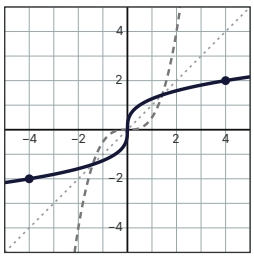
B. SHOW CORRECT WORK (1 POINT)

$(x, y) \to (y, x)$: $(2, 4) \to (4, 2)$, $(-2, -4) \to (-4, -2)$

Reflect f in the line $y = x$

Algebra: $x = \frac{y^3}{2}$, $y^3 = 2x$

$f^{-1}(x) = \sqrt[3]{2x}$



Dashed: $f(x)$ and $y = x$. Solid: $f^{-1}(x)$.

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points

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9. Piecewise: boundary points

ORIGINAL PROBLEM

Graph f and find $f(1)$.
$f(x) = x + 2$ for $x < 1$
$f(x) = x^2 - 2$ for $x \geq 1$

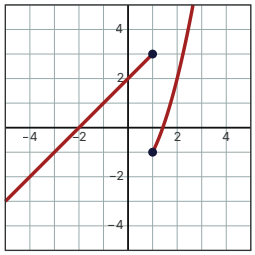
INCORRECT WORKED SOLUTION

Left piece at $x = 1$: $1 + 2 = 3$

Right piece at $x = 1$: $1^2 - 2 = -1$

Closed dot at (1, 3) and closed dot at (1, −1)

So $f(1) = 3$ and $f(1) = -1$.



Student's graph.

A. EXPLAIN THE MISTAKE (1 POINT)

$x < 1$ is strict, so (1, 3) is an OPEN dot; only $x \geq 1$ includes 1, so (1, −1) is closed. A function has ONE output per x; the student's graph fails the vertical line test.

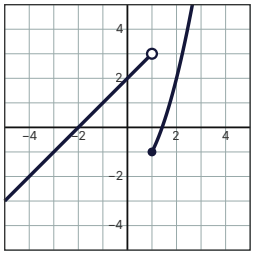
B. SHOW CORRECT WORK (1 POINT)

$x < 1$ is strict: open dot at (1, 3)

$x \geq 1$ includes 1: closed dot at (1, −1)

$f(1) = 1^2 - 2 = -1$

One output for $x = 1$, so f passes the vertical line test.



Open dot: not included. Closed dot: included.

10. Order of transformations

ORIGINAL PROBLEM

The point (3, −2) is on the graph of $y = f(x)$. Find the corresponding point on $y = -2f(x + 1) + 3$.

INCORRECT WORKED SOLUTION

x: $x + 1 = 3$, so $x = 2$

y: $-2 + 3 = 1$

then $1 \cdot (-2) = -2$

Point: (2, −2)

A. EXPLAIN THE MISTAKE (1 POINT)

Outside the function, follow the order of operations: multiply by −2 (reflect and stretch) BEFORE adding 3 (the vertical shift). The student added 3 first. The x-coordinate was correct.

B. SHOW CORRECT WORK (1 POINT)

x: $x + 1 = 3$, so $x = 2$ (left 1)

y: $(-2)(-2) = 4$ first

then $4 + 3 = 7$

Point: (2, 7)

Scoring: Explanation = 1 point • Correct work/answer = 1 point • Total = 20 points